

# Embeddings contextualizados

Materia: Procesamiento de lenguaje natural  
Blanca Vázquez

# Embeddings contextuales

---

A diferencia de las estrategias anteriores, que generan un único vector para cada palabra independientemente del contexto, los embeddings contextuales generan **diferentes vectores para la misma palabra** dependiendo de su uso en la oración.

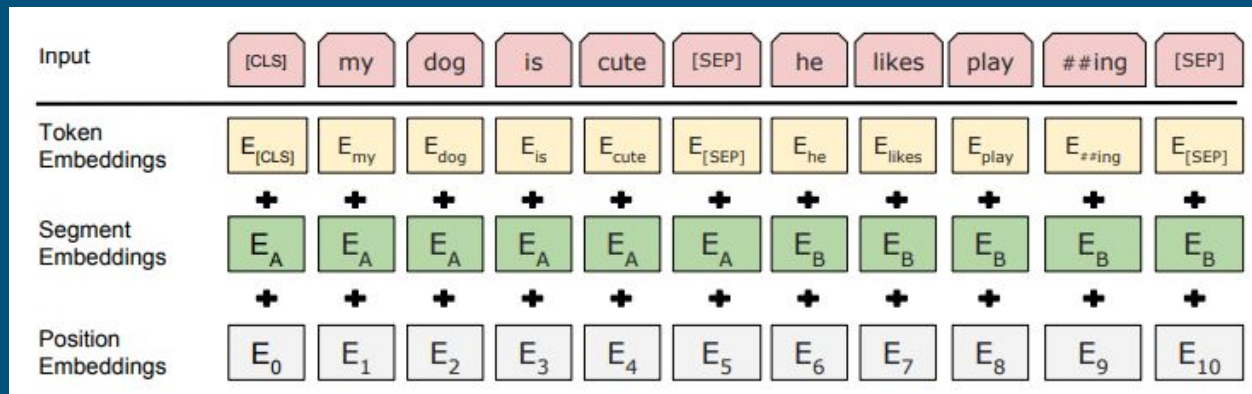
Ejemplo, la palabra "banco" tendrá un vector diferente en la oración

- "Me senté en el **banco** del parque" que en
- "El **banco** aprobó mi solicitud de crédito".

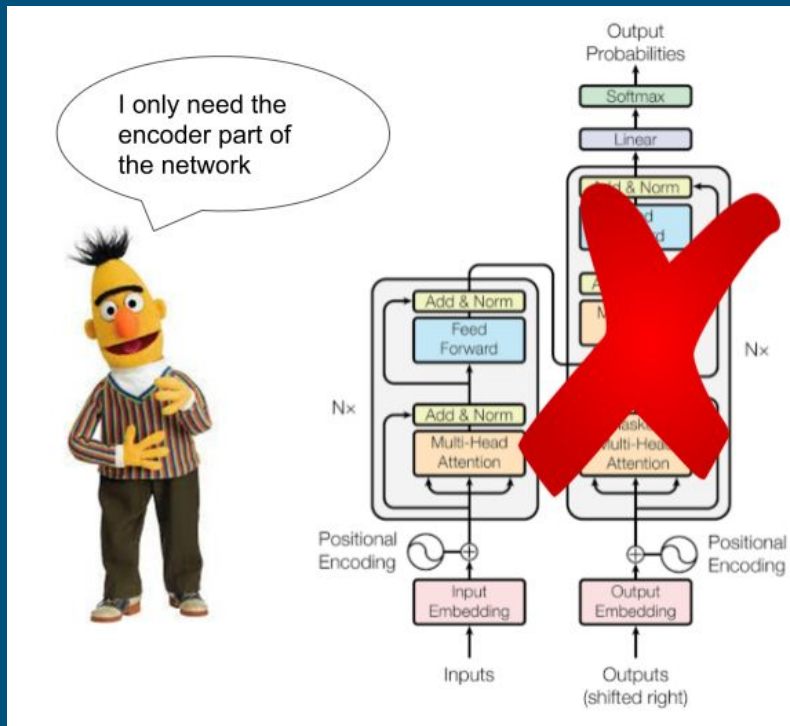
Esta variabilidad se logra entrenando el modelo con corpus de texto extensos de forma **bidireccional**, es decir, considerando no sólo las palabras que preceden a la palabra objetivo, sino también las que la siguen.

# BERT (Bidirectional Encoder Representations from Transformers)

BERT es un modelo de código abierto propuesto por Devlin, Chang, Lee y Toutanova miembros de Google en el 2018. Está basado en mecanismos de atención y son bidireccionales.



# Diferencia entre Transformers y BERT



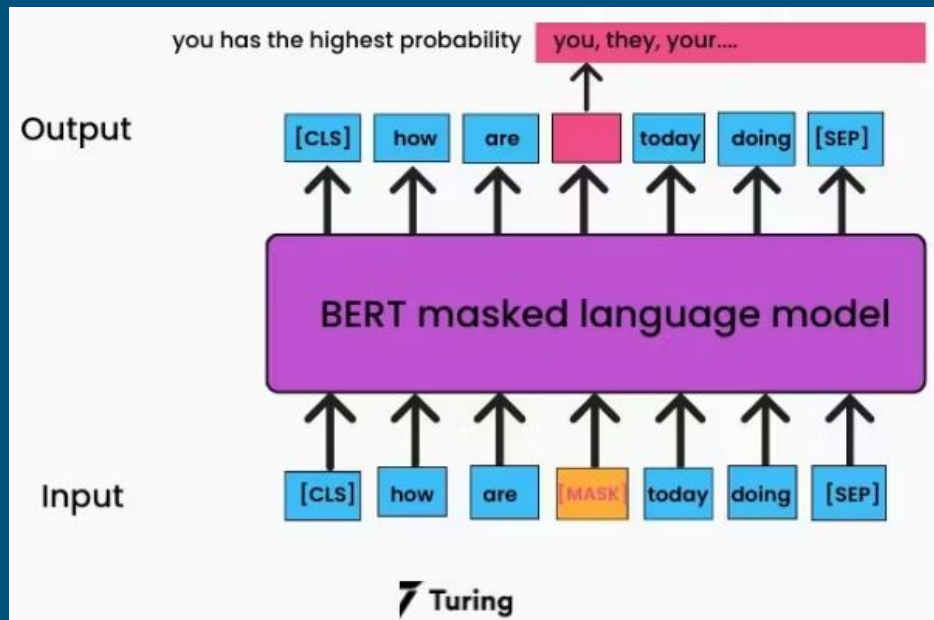
# ¿Por qué usar únicamente el encoder?

---

- BERT presta un mayor énfasis en comprender las secuencias de entrada en lugar de generar secuencias de salida.
- Usando únicamente el encoder, BERT es capaz de codificar la información semántica y sintáctica en los embeddings.
- La salida final de BERT es un embedding, no es una salida textual.
- BERT es capaz de “observar” todas las palabras de una oración simultáneamente (bidireccional).

# Masked Language Model (MLM)

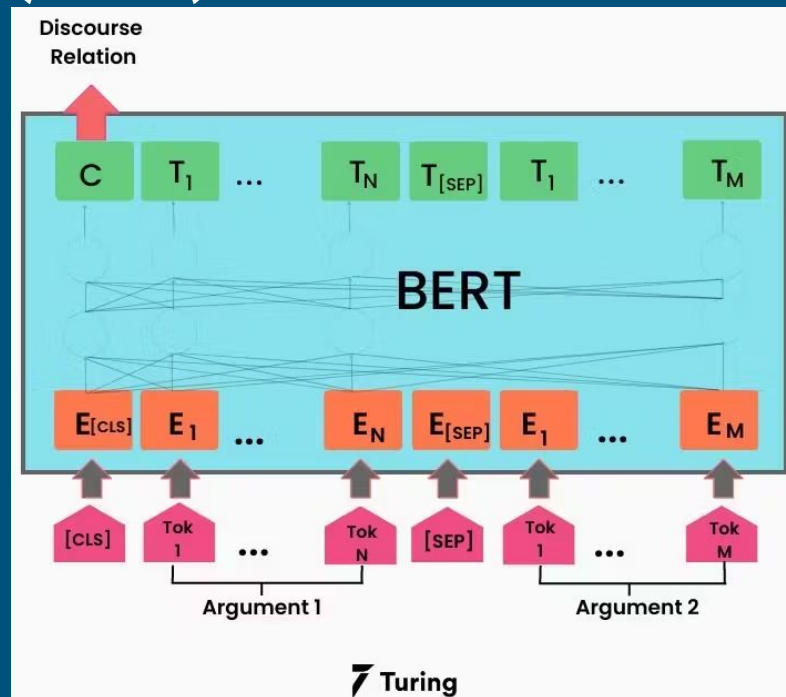
1. Se añade una **capa de clasificación** a la salida del codificador para predecir las palabras enmascaradas.
2. Los vectores de salida se multiplican por la matriz de embeddings, transformándolos en la dimensión del **vocabulario**. Esto para alinear las representaciones predichas con el espacio del vocabulario.
3. Para calcular la probabilidad de cada palabra, se usa una función **SoftMax**.
4. La **función de pérdida** utilizada durante el entrenamiento solo considera la predicción de los valores enmascarados.
5. El modelo se penaliza por la desviación entre sus predicciones y los valores reales de las palabras enmascaradas.



Durante el entrenamiento, una selección aleatoria del **15%** de las palabras tokenizadas se enmascaran.

# Next Sentence Prediction (NSP)

1. Durante el entrenamiento, BERT predice si la segunda oración sigue a la primera en el documento original.
2. BERT transforma la salida del token [CLS] en un vector con forma de  $2 \times 1$  mediante una capa de clasificación.
3. A continuación, calcula la probabilidad de que la segunda oración siga a la primera mediante SoftMax.



El 50% de los pares de entrada tienen la segunda oración como la oración subsiguiente en el documento original, y el otro 50 % tiene una oración elegida aleatoriamente.

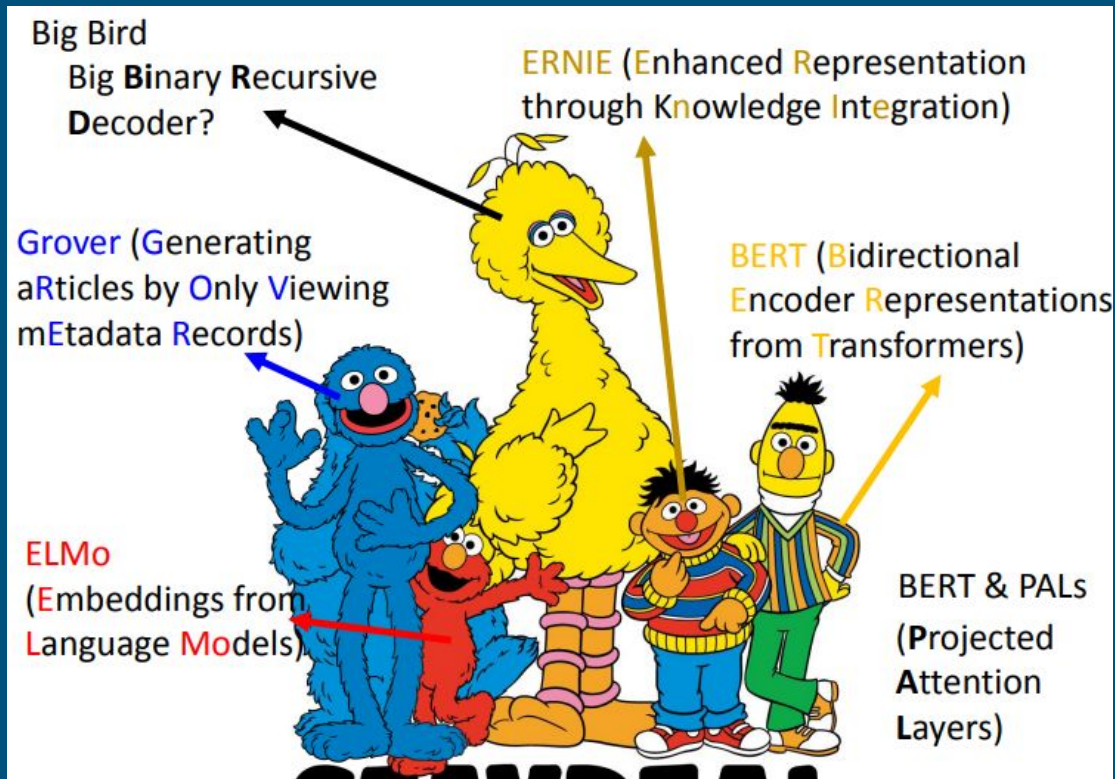
# Importante

---

- Durante el entrenamiento, el MLM enmascarado y la predicción de la siguiente oración se entrenan **conjuntamente**.
- El modelo busca **minimizar la función de pérdida combinada** de ambos, lo que resulta en un modelo lingüístico robusto con mayor capacidad para comprender el contexto dentro de las oraciones y las relaciones entre ellas.



# Variantes y adaptaciones del BERT



# GLUE score

GLUE por las siglas en inglés de *General Language Understanding Evaluation* es una colección de recursos para entrenar, evaluar y analizar sistemas de comprensión del lenguaje natural.

GLUE consta de **nueve conjuntos** de tareas “difíciles y diversas” diseñados para probar la comprensión del lenguaje de un modelo.

| Acrónimo | Descripción                                       |
|----------|---------------------------------------------------|
| COLA     | The Corpus of Linguistic Acceptability            |
| SST-2    | The Stanford Sentiment Treebank                   |
| MRPC     | The Microsoft Research Paraphrase Corpus          |
| QQP      | The Quora Question Pairs                          |
| STS-B    | The Semantic Textual Similarity Benchmark         |
| MNLI     | The Multi-Genre Natural Language Inference Corpus |
| QNLI     | The Stanford Question Answering Dataset           |
| RTE      | The Recognizing Textual Entailment                |
| WNLI     | The Winograd Schema Challenge                     |

# GLUE score

| Corpus                          | Train | Test        | Task                | Metrics                      | Domain              |
|---------------------------------|-------|-------------|---------------------|------------------------------|---------------------|
| Single-Sentence Tasks           |       |             |                     |                              |                     |
| CoLA                            | 8.5k  | <b>1k</b>   | acceptability       | Matthews corr.               | misc.               |
| SST-2                           | 67k   | 1.8k        | sentiment           | acc.                         | movie reviews       |
| Similarity and Paraphrase Tasks |       |             |                     |                              |                     |
| MRPC                            | 3.7k  | 1.7k        | paraphrase          | acc./F1                      | news                |
| STS-B                           | 7k    | 1.4k        | sentence similarity | Pearson/Spearman corr.       | misc.               |
| QQP                             | 364k  | <b>391k</b> | paraphrase          | acc./F1                      | social QA questions |
| Inference Tasks                 |       |             |                     |                              |                     |
| MNLI                            | 393k  | <b>20k</b>  | NLI                 | matched acc./mismatched acc. | misc.               |
| QNLI                            | 105k  | 5.4k        | QA/NLI              | acc.                         | Wikipedia           |
| RTE                             | 2.5k  | 3k          | NLI                 | acc.                         | news, Wikipedia     |
| WNLI                            | 634   | <b>146</b>  | coreference/NLI     | acc.                         | fiction books       |

Table 1: Task descriptions and statistics. All tasks are single sentence or sentence pair classification, except STS-B, which is a regression task. MNLI has three classes; all other classification tasks have two. Test sets shown in bold use labels that have never been made public in any form.

| Model                                      | Avg         | Single Sentence |             | Similarity and Paraphrase |                  |                  | Natural Language Inference |             |             |             |
|--------------------------------------------|-------------|-----------------|-------------|---------------------------|------------------|------------------|----------------------------|-------------|-------------|-------------|
|                                            |             | CoLA            | SST-2       | MRPC                      | QQP              | STS-B            | MNLI                       | QNLI        | RTE         | WNLI        |
| Single-Task Training                       |             |                 |             |                           |                  |                  |                            |             |             |             |
| BiLSTM                                     | 63.9        | 15.7            | 85.9        | 69.3/79.4                 | 81.7/61.4        | 66.0/62.8        | 70.3/70.8                  | 75.7        | 52.8        | <b>65.1</b> |
| +ELMo                                      | 66.4        | <b>35.0</b>     | <u>90.2</u> | 69.0/80.8                 | 85.7/65.6        | 64.0/60.2        | 72.9/73.4                  | 71.7        | 50.1        | <b>65.1</b> |
| +CoVe                                      | 64.0        | <u>14.5</u>     | 88.5        | <u>73.4/81.4</u>          | 83.3/59.4        | <u>67.2/64.1</u> | 64.5/64.8                  | 75.4        | <u>53.5</u> | <b>65.1</b> |
| +Attn                                      | 63.9        | 15.7            | 85.9        | 68.5/80.3                 | 83.5/62.9        | 59.3/55.8        | 74.2/73.8                  | <u>77.2</u> | 51.9        | <b>65.1</b> |
| +Attn, ELMo                                | <u>66.5</u> | <b>35.0</b>     | <u>90.2</u> | 68.8/80.2                 | <b>86.5/66.1</b> | 55.5/52.5        | <b>76.9/76.7</b>           | 76.7        | 50.4        | <b>65.1</b> |
| +Attn, CoVe                                | 63.2        | <u>14.5</u>     | 88.5        | 68.6/79.7                 | 84.1/60.1        | 57.2/53.6        | 71.6/71.5                  | 74.5        | 52.7        | <b>65.1</b> |
| Multi-Task Training                        |             |                 |             |                           |                  |                  |                            |             |             |             |
| BiLSTM                                     | 64.2        | 11.6            | 82.8        | 74.3/81.8                 | 84.2/62.5        | 70.3/67.8        | 65.4/66.1                  | 74.6        | 57.4        | <b>65.1</b> |
| +ELMo                                      | 67.7        | 32.1            | 89.3        | <b>78.0/84.7</b>          | 82.6/61.1        | 67.2/67.9        | 70.3/67.8                  | 75.5        | 57.4        | <b>65.1</b> |
| +CoVe                                      | 62.9        | 18.5            | 81.9        | 71.5/78.7                 | <u>84.9/60.6</u> | 64.4/62.7        | 65.4/65.7                  | 70.8        | 52.7        | <b>65.1</b> |
| +Attn                                      | 65.6        | 18.6            | 83.0        | 76.2/83.9                 | 82.4/60.1        | 72.8/70.5        | 67.6/68.3                  | 74.3        | 58.4        | <b>65.1</b> |
| +Attn, ELMo                                | <b>70.0</b> | <u>33.6</u>     | <b>90.4</b> | <b>78.0/84.4</b>          | <u>84.3/63.1</u> | <u>74.2/72.3</u> | <u>74.1/74.5</u>           | <b>79.8</b> | <u>58.9</u> | <b>65.1</b> |
| +Attn, CoVe                                | 63.1        | 8.3             | 80.7        | 71.8/80.0                 | 83.4/60.5        | 69.8/68.4        | 68.1/68.6                  | 72.9        | 56.0        | <b>65.1</b> |
| Pre-Trained Sentence Representation Models |             |                 |             |                           |                  |                  |                            |             |             |             |
| CBoW                                       | 58.9        | 0.0             | 80.0        | 73.4/81.5                 | 79.1/51.4        | 61.2/58.7        | 56.0/56.4                  | 72.1        | 54.1        | <b>65.1</b> |
| Skip-Thought                               | 61.3        | 0.0             | 81.8        | 71.7/80.8                 | 82.2/56.4        | 71.8/69.7        | 62.9/62.8                  | 72.9        | 53.1        | <b>65.1</b> |
| InferSent                                  | 63.9        | 4.5             | <u>85.1</u> | 74.1/81.2                 | 81.7/59.1        | 75.9/75.3        | 66.1/65.7                  | 72.7        | 58.0        | <b>65.1</b> |
| DisSent                                    | 62.0        | 4.9             | 83.7        | 74.1/81.7                 | 82.6/59.5        | 66.1/64.8        | 58.7/59.1                  | 73.9        | 56.4        | <b>65.1</b> |
| GenSen                                     | <u>66.2</u> | <u>7.7</u>      | 83.1        | <u>76.6/83.0</u>          | <u>82.9/59.8</u> | <b>79.3/79.2</b> | <u>71.4/71.3</u>           | <u>78.6</u> | <b>59.2</b> | <b>65.1</b> |

Table 4: Baseline performance on the GLUE task test sets. For MNLI, we report accuracy on the matched and mismatched test sets. For MRPC and Quora, we report accuracy and F1. For STS-B, we report Pearson and Spearman correlation. For CoLA, we report Matthews correlation. For all other tasks we report accuracy. All values are scaled by 100. A similar table is presented on the online platform.

# Sitio oficial de GLUE

---

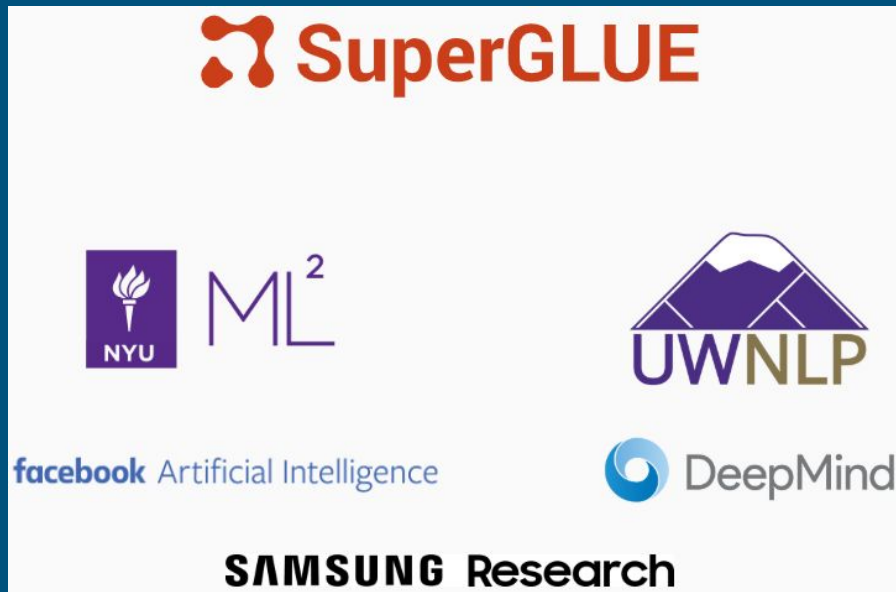


<https://gluebenchmark.com/>

# Super GLUE

---

Punto de referencia inspirado en GLUE con un nuevo conjunto de tareas de comprensión del lenguaje más difíciles, recursos mejorados y una nueva tabla de clasificación pública.



<https://super.gluebenchmark.com/>

# Super GLUE tareas

---

| Acrónimo | Descripción                                              |
|----------|----------------------------------------------------------|
| BoolQ    | Boolean Questions                                        |
| CB       | CommitmentBank                                           |
| COPA     | Choice of Plausible Alternatives                         |
| MultiRC  | Multi-Sentence Reading Comprehension                     |
| ReCoRD   | Reading Comprehension with Commonsense Reasoning Dataset |
| RTE      | Recognizing Textual Entailment                           |
| WiC      | Word-in-Context                                          |
| WSC      | Winograd Schema Challenge                                |



# Super GLUE tareas

Table 1: The tasks included in SuperGLUE. *WSD* stands for word sense disambiguation, *NLI* is natural language inference, *coref.* is coreference resolution, and *QA* is question answering. For MultiRC, we list the number of total answers for 456/83/166 train/dev/test questions. The metrics for MultiRC are binary F1 on all answer-options and exact match.

| Corpus  | Train | Dev  | Test | Task   | Metrics             | Text Sources                    |
|---------|-------|------|------|--------|---------------------|---------------------------------|
| BoolQ   | 9427  | 3270 | 3245 | QA     | acc.                | Google queries, Wikipedia       |
| CB      | 250   | 57   | 250  | NLI    | acc./F1             | various                         |
| COPA    | 400   | 100  | 500  | QA     | acc.                | blogs, photography encyclopedia |
| MultiRC | 5100  | 953  | 1800 | QA     | F1 <sub>a</sub> /EM | various                         |
| ReCoRD  | 101k  | 10k  | 10k  | QA     | F1/EM               | news (CNN, Daily Mail)          |
| RTE     | 2500  | 278  | 300  | NLI    | acc.                | news, Wikipedia                 |
| WiC     | 6000  | 638  | 1400 | WSD    | acc.                | WordNet, VerbNet, Wiktionary    |
| WSC     | 554   | 104  | 146  | coref. | acc.                | fiction books                   |



# Tabla de clasificación

| Rank Name |                            | Model            | URL                                                                                 | Score | CoLA | SST-2 | MRPC      | STS-B     | QQP       | MNLI-m | MNLI-mm | QNLI | RTE  | WNLI |
|-----------|----------------------------|------------------|-------------------------------------------------------------------------------------|-------|------|-------|-----------|-----------|-----------|--------|---------|------|------|------|
| 1         | Microsoft Alexander v-team | Turing ULR v6    |  | 91.3  | 73.3 | 97.5  | 94.2/92.3 | 93.5/93.1 | 76.4/90.9 | 92.5   | 92.1    | 96.7 | 93.6 | 97.9 |
| 2         | JDExplore d-team           | Vega v1          |                                                                                     | 91.3  | 73.8 | 97.9  | 94.5/92.6 | 93.5/93.1 | 76.7/91.1 | 92.1   | 91.9    | 96.7 | 92.4 | 97.9 |
| 3         | Microsoft Alexander v-team | Turing NLR v5    |  | 91.2  | 72.6 | 97.6  | 93.8/91.7 | 93.7/93.3 | 76.4/91.1 | 92.6   | 92.4    | 97.9 | 94.1 | 95.9 |
| 4         | DIRL Team                  | DeBERTa + CLEVER |                                                                                     | 91.1  | 74.7 | 97.6  | 93.3/91.1 | 93.4/93.1 | 76.5/91.0 | 92.1   | 91.8    | 96.7 | 93.2 | 96.6 |
| 5         | ERNIE Team - Baidu         | ERNIE            |  | 91.1  | 75.5 | 97.8  | 93.9/91.8 | 93.0/92.6 | 75.2/90.9 | 92.3   | 91.7    | 97.3 | 92.6 | 95.9 |

<https://gluebenchmark.com/leaderboard/>

# Large Language Model

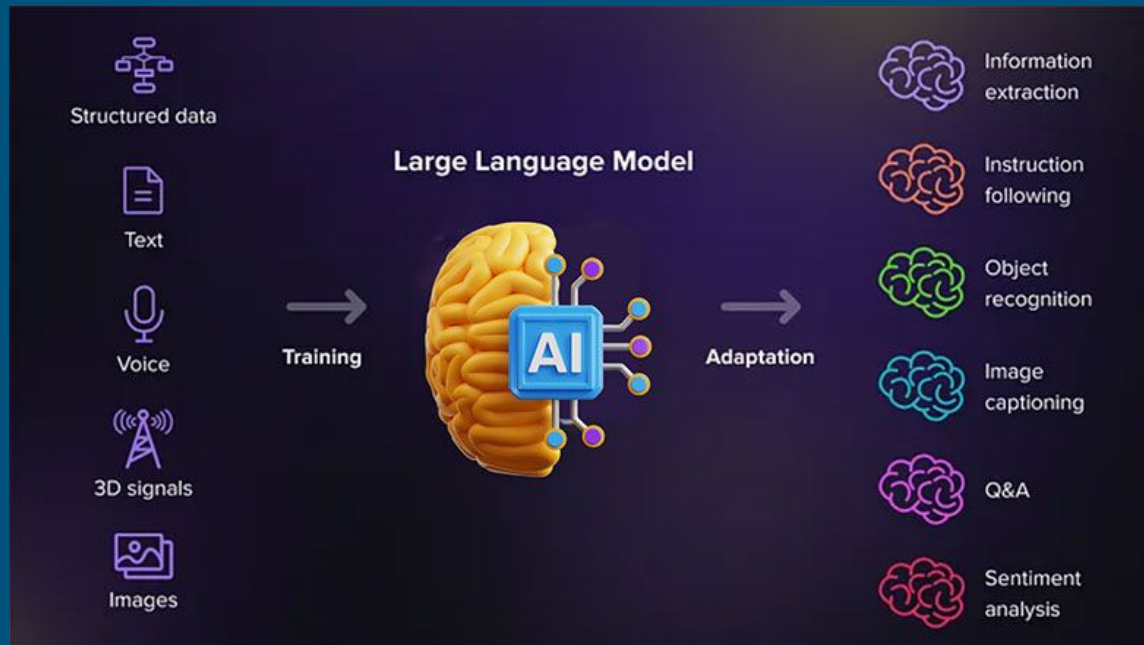
Materia: Procesamiento de lenguaje natural  
Blanca Vázquez

# Large Language Model

Son modelos de IA, específicamente redes neuronales, entrenados con conjuntos masivos de datos de texto.

Están diseñados para comprender, generar y manipular el lenguaje humano, lo que les permite realizar tareas como la generación de texto, la traducción, la respuesta a preguntas y la escritura de código.

Los LLM son una herramienta poderosa en el PLN y se utilizan cada vez más en diversas aplicaciones.



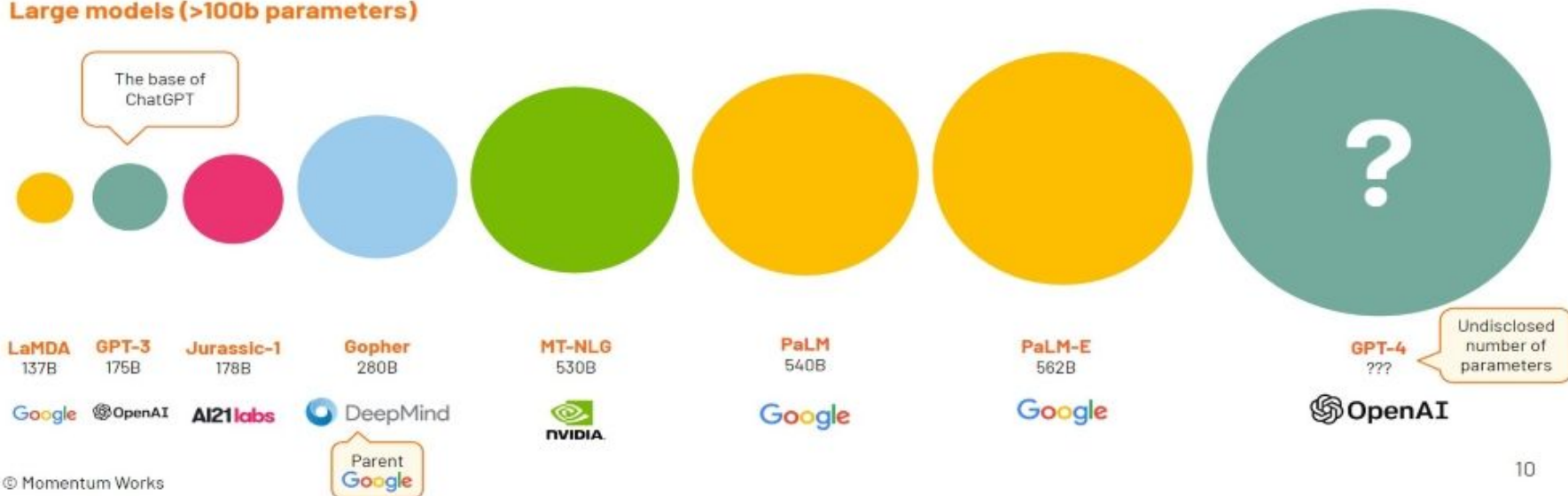
## Large Language Models are becoming very large indeed



### Small models (<= 100b parameters)



### Large models (>100b parameters)



# The most popular Large Language Models nowadays are:

As of mid-2025, the most popular and widely used large language models (LLMs) include:

## General-Purpose LLMs

These are the most prominent and widely adopted:

1. **GPT-4 / GPT-4o** – *OpenAI*
  - GPT-4o ("omni") is the latest flagship model from OpenAI.
  - Multimodal: supports text, image, audio, and video (in Sora).
  - Available through ChatGPT, API, and integrations like Microsoft Copilot.
2. **Claude 3 Family (Haiku, Sonnet, Opus)** – *Anthropic*
  - Known for safety and long-context reasoning (up to 200K tokens).
  - Opus is the most capable; Haiku is the fastest/lightest.
  - Integrated in services like Slack and Notion.
3. **Gemini 1.5** – *Google DeepMind*
  - Multimodal and high-context window (up to 1M tokens).
  - Powers Google's Bard (now also branded as Gemini).
4. **Mistral & Mixtral Models** – *Mistral AI*
  - Open-weight models; fast and efficient.
  - Mixtral is a mixture-of-experts (MoE) model.
5. **LLaMA 3 (Large Language Model Meta AI)** – *Meta*
  - Open-source models: LLaMA 3 8B and 70B available.
  - Used in many custom applications and platforms.
6. **Command R+** – *Cohere*
  - Optimized for retrieval-augmented generation (RAG).
  - Efficient in enterprise and knowledge-based tasks.

## Specialized or Open-Source LLMs

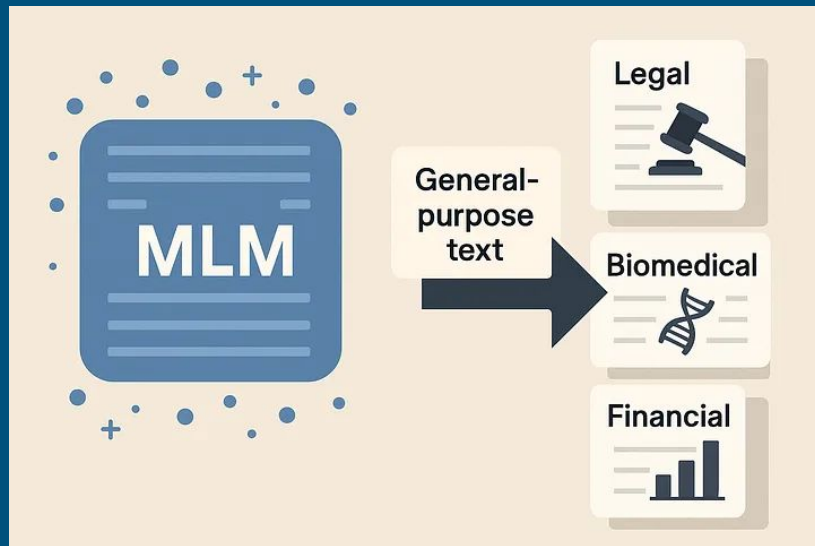
Often used in academic, research, or fine-tuned applications:

- **Phi-3** – *Microsoft*  
Small, performant models for local use.
- **OpenHermes, Orca, Zephyr, MPT, Falcon** – *Various*  
Community-developed or research-tuned models.
- **xAI's Grok (Grok-1)** – *Elon Musk's xAI*  
Used within X (formerly Twitter), partially open-sourced.

Respuestas generadas por ChatGPT

# Aplicaciones

- Análisis de sentimientos
- Traducción de idiomas
- Respuesta a preguntas
- Búsqueda de Google
- Resumen de texto
- Coincidencia y recuperación de texto
- Resaltado de párrafos



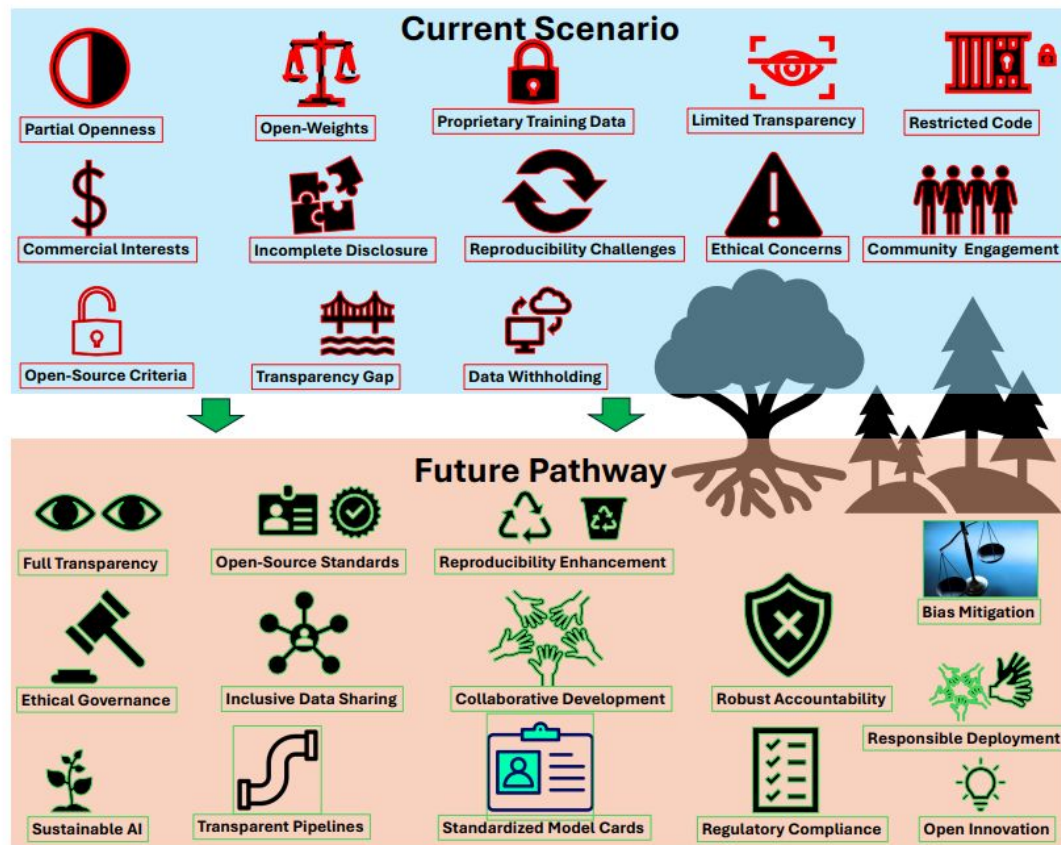


Figure 6: Illustrating key dimensions of transparency and accessibility practices in LLMs and outlines of future pathways. The upper panel displays current practices, such as partial openness, proprietary training data, limited transparency, and restricted code disclosure. In contrast, the lower panel indicates a future pathway toward full transparency, and ethical governance, inclusive data sharing, enhanced reproducibility, and sustainability



| Model                                           | Carbon Emissions<br>(Metric Tons CO <sub>2</sub> )<br>during<br>Pre-training | Equivalent<br>Number of Trees |
|-------------------------------------------------|------------------------------------------------------------------------------|-------------------------------|
| GPT-3 Brown et al. (2020)                       | 552                                                                          | 25,091                        |
| LLaMA 2 70B Touvron et al. (2023) <sup>9</sup>  | 291.42                                                                       | 13,247                        |
| Llama 3.1 70B Dubey et al. (2024) <sup>10</sup> | 2040                                                                         | 92,727                        |
| Llama 3.2 1B Dubey et al. (2024)                | 71                                                                           | 3,227                         |
| Llama 3.2 3B Dubey et al. (2024)                | 133                                                                          | 6,045                         |



# LLM Leaderboard - Comparison of over 100 AI models from OpenAI, Google, DeepSeek & others

Filters

Frontier Models:

AllFrontier Models

Open Weights:

AllOpen SourceProprietary

Size Class:

AllTinySmallMediumLarge

Reasoning:

AllReasoningNon Reasoning

|               | FEATURES        | INTELLIGENCE   |                                        | PRICE                 | OUTPUT TOKENS/S | LATENCY                |                  |
|---------------|-----------------|----------------|----------------------------------------|-----------------------|-----------------|------------------------|------------------|
| MODEL         | CREATOR         | CONTEXT WINDOW | ARTIFICIAL ANALYSIS INTELLIGENCE INDEX | BLENDED USD/1M Tokens | MEDIAN Tokens/s | MEDIAN First Chunk (s) | FURTHER ANALYSIS |
| Llama 65B     | Meta            | 2k             | 11                                     | \$0.00                |                 |                        | ModelProviders   |
| Arctic        | snowflake       | 4k             | 22                                     | \$0.00                |                 |                        | ModelProviders   |
| Solar Mini    | upstage         | 4k             | 33                                     | \$0.15                | 86.6            | 1.07                   | ModelProviders   |
| GPT-3.5 Turbo | OpenAI          | 4k             | 23                                     | \$0.75                | 112.4           | 0.43                   | ModelProviders   |
| Phi-3 Mini    | Microsoft Azure | 4k             | 22                                     | \$0.00                |                 |                        | ModelProviders   |

<https://artificialanalysis.ai/leaderboards/models>